Which Covariance Forecast Survives a Century?

Out-of-sample minimum-variance evaluation with Model Confidence Set inference

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Abstract We compare 9 covariance estimators, from the sample covariance to nonlinear shrinkage and dynamic conditional correlation, by the realized risk of the global minimum variance portfolios they produce on daily US portfolio returns from the Kenneth R. French Data Library. On 49 industries over 1930 to 2026 with a 252-day window, LW constant-corr attains the lowest annualized out-of-sample volatility, 9.51% against 9.78% for the sample covariance. The 90% Model Confidence Set retains 3 of 8 models. A known-covariance simulation reproduces the same estimators in a setting where the true risk of every portfolio is observable.

1 Introduction

The covariance matrix is the input every risk model and every mean-variance optimizer shares. Its textbook estimator, the sample covariance, is unbiased entry by entry but its eigenvalues are systematically too dispersed: large ones are overestimated and small ones underestimated, and the error grows with the ratio of assets N to observations T. A minimum variance optimizer inverts the matrix, so it loads precisely on the directions whose variance is most underestimated. The literature offers several remedies: shrinkage toward a structured target, factor structure, and conditional models that track changing volatility and correlation.

This study asks a narrow, testable question. Holding the portfolio construction fixed at the global minimum variance (GMV) portfolio, which uses no expected returns, which estimator produces the lowest realized risk out of sample, how does the ranking change with N/T, and which estimators cannot be distinguished statistically?

2 Data

Returns are daily value-weighted portfolio returns from the Kenneth R. French Data Library: 10 industries, 30 industries, 49 industries and 100 size-B/M portfolios. Factor returns for the observable factor models come from the Fama-French three-factor file and, from 1963-07-01, the five-factor and momentum files. Every file is downloaded over HTTPS, its SHA-256 recorded in a manifest, and parsed structurally rather than by line position. Portfolios with no constituents carry the library's missing-value sentinel; an asset enters an estimation window only if it has a return on every day of that window.

Table 1: Data coverage as parsed.

File	First day	Last day	Days	Series	Missing share	All series from	Retrieved
ff3	1926-07-01	2026-09-30	26338	4	0.000	1926-07-01	2026-11-03
ff5	1963-07-01	2026-09-30	15918	6	0.000	1963-07-01	2026-11-03
ind10	1926-07-01	2026-09-30	26338	10	0.000	1926-07-01	2026-11-03
ind30	1926-07-01	2026-09-30	26338	30	0.000	1926-07-01	2026-11-03
ind49	1926-07-01	2026-09-30	26338	49	0.052	1969-07-01	2026-11-03
me_bm100	1926-07-01	2026-09-30	26338	100	0.013	1945-07-02	2026-11-03
mom	1926-11-03	2026-09-30	26237	1	0.000	1926-11-03	2026-11-03

3 Method

3.1 Evaluation design

At the last trading day of each month t, every model is fitted on the trailing window of T trading days, with T set to 126, 252, 504 and 1260 and forecasts the average daily covariance H_t over the next month. The GMV weights

$$w_t = \frac{H_t^{-1}\mathbf{1}}{\mathbf{1}^\top H_t^{-1}\mathbf{1}}$$

are held fixed through the month, and the long-only variant solves $\min_w w^\top H_t w$ subject to $w \geq 0$, $\mathbf{1}^\top w = 1$ by accelerated projected gradient. Membership of the estimation window is decided from the window alone, never from the holding period, and the test suite verifies that perturbing returns after a rebalance date leaves that date's weights unchanged.

3.2 Losses

The economic loss is the realized variance of the GMV portfolio. Two statistical losses compare H_t with the realized covariance proxy $\Sigma_t = D^{-1}\sum_d r_d r_d^\top$ over the D days of the month: the squared Frobenius distance and the multivariate QLIKE,

$$L(\Sigma_t, H_t) = \log|H_t| + \operatorname{tr}(H_t^{-1}\Sigma_t).$$

Both rank forecasts consistently when the proxy is conditionally unbiased but noisy (Laurent, Rombouts and Violante, 2013), which matters because Σ_t is singular whenever $N > D$.

3.3 Estimators

Linear shrinkage (Ledoit and Wolf, 2004) forms $\delta F + (1-\delta)S$ with F the constant-correlation target and δ the estimated optimal intensity.

Nonlinear shrinkage (Ledoit and Wolf, 2020) keeps the sample eigenvectors u_i and replaces each sample eigenvalue λ_i by

$$\tilde{d}_i = \frac{\lambda_i}{\left(\pi \frac{p}{n}\lambda_i \tilde{f}(\lambda_i)\right)^2 + \left(1 - \frac{p}{n} - \pi\frac{p}{n}\lambda_i \mathcal{H}_{\tilde{f}}(\lambda_i)\right)^2},$$

where $\tilde{f}$ is an Epanechnikov kernel estimate of the limiting spectral density and $\mathcal{H}_{\tilde{f}}$ its Hilbert transform, with a separate formula for the null space when $p > n$.

Factor models set $H = B\Sigma_f B^\top + D$ with D diagonal, using either observable Fama-French factors or principal components, the number of components chosen inside each window by the Bai and Ng (2002) criterion IC_{p2}.

DCC fits a variance-targeted GARCH(1,1) to each asset and the scalar correlation recursion

$$Q_t = (1 - a - b)\, C + a\, z_{t-1} z_{t-1}^{\top} + b\, Q_{t-1}, \qquad R_t = \operatorname{diag}(Q_t)^{-1/2} Q_t \operatorname{diag}(Q_t)^{-1/2},$$

estimated by the composite likelihood over contiguous pairs (Engle, Shephard and Sheppard, 2008). DCC-NL replaces the target C by its nonlinear-shrinkage estimate (Engle, Ledoit and Wolf, 2019). Multi-step forecasts use exact GARCH mean reversion and the Engle-Sheppard approximation for correlations.

3.4 Model Confidence Set

Statistical separation uses the Model Confidence Set (Hansen, Lunde and Nason, 2011) with the $T_{\max}$ statistic, elimination of the model with the largest standardized excess loss, and the stationary bootstrap with mean block length 6.0 months and 2000 replications. The set at level $\alpha = 0.1$ contains every model whose MCS p-value is at least α. A comparison only includes models defined in every month of its sample; exclusions are reported, not imputed.

4 Results

Table 2: Full-sample out-of-sample performance, 49 industries, 252-day window, 1152 months.

Model	GMV vol (%)	Long-only vol (%)	Turnover (%/month)	QLIKE
LW constant-corr	9.51	11.10	73.0	−401.287
LW nonlinear	9.55	11.16	71.6	−402.124
DCC-NL	9.62	11.23	145.4	−402.500
Sample	9.78	11.14	98.1	−397.592
DCC-GARCH	9.95	11.27	182.1	−397.790
FF3 factor	9.99	11.15	41.9	−401.925
PCA factor	10.18	11.19	43.7	−400.832
EWMA	13.94	12.00	571.4	−158.985

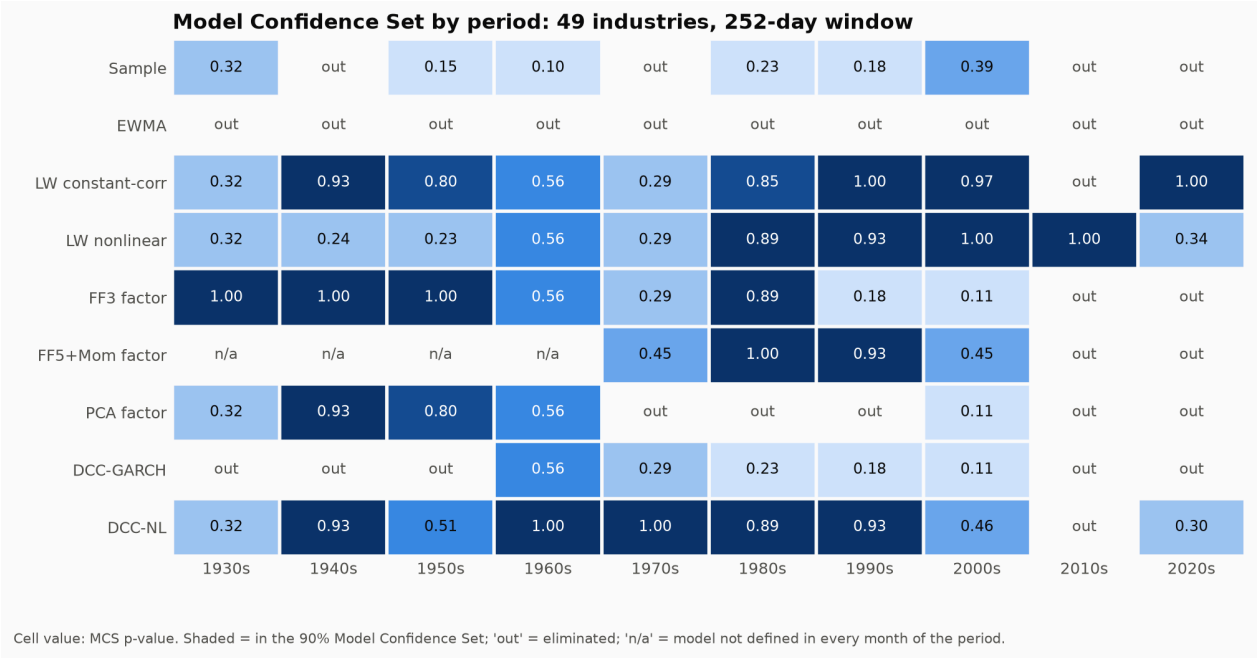


Figure 1: MCS p-values for the GMV variance loss. Shaded cells are inside the 90% set.

Over the full sample for 10 industries, the 90% set under the GMV variance loss contains DCC-GARCH, DCC-NL, FF3 factor, LW constant-corr, LW nonlinear and Sample. Over the full sample for 30 industries, the 90% set

under the GMV variance loss contains DCC-NL, LW constant-corr and LW nonlinear. Over the full sample for 49 industries, the 90% set under the GMV variance loss contains DCC-NL, LW constant-corr and LW nonlinear. Over the full sample for 100 size-B/M portfolios, the 90% set under the GMV variance loss contains DCC-NL, FF3 factor, LW constant-corr and LW nonlinear.

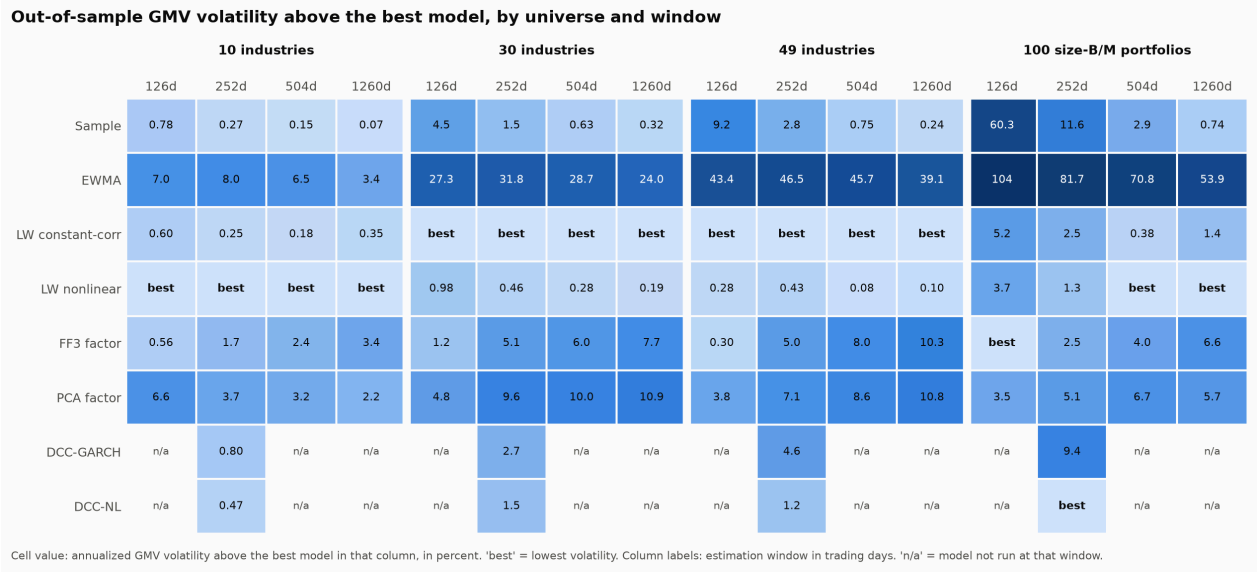


Figure 2: Excess GMV volatility over the best model, by universe and estimation window.

Annualized volatility levels for every universe, window and model are in `reports/tables/sweep_vol.md` in the repository.

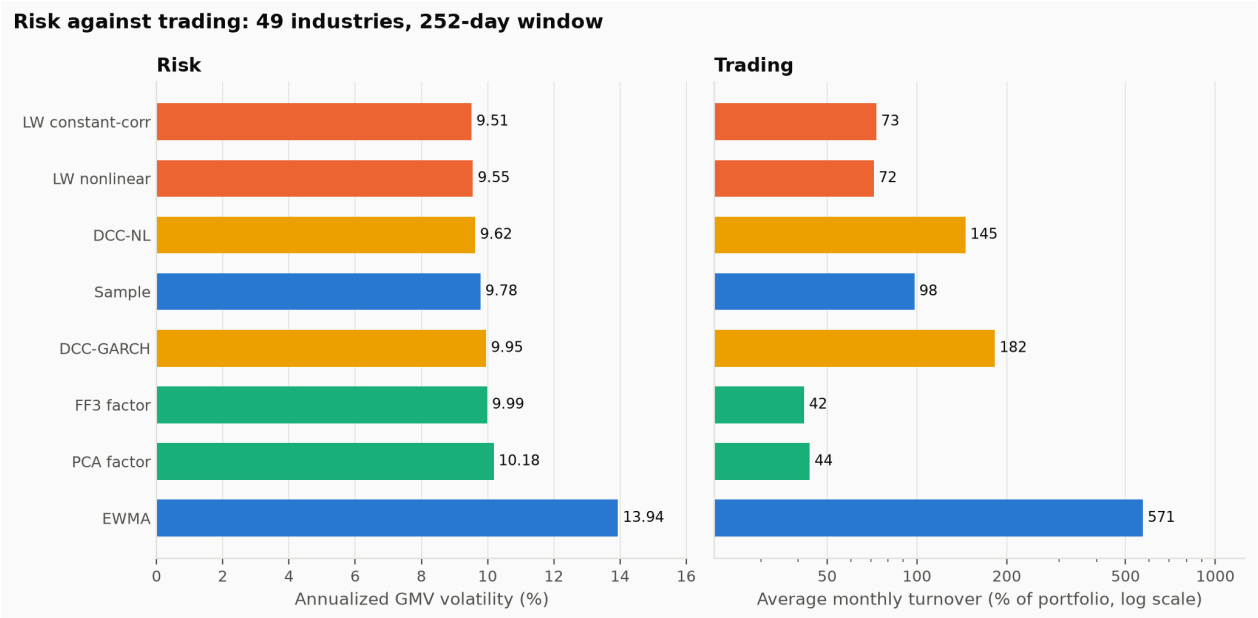


Figure 3: Average monthly turnover against realized GMV volatility.

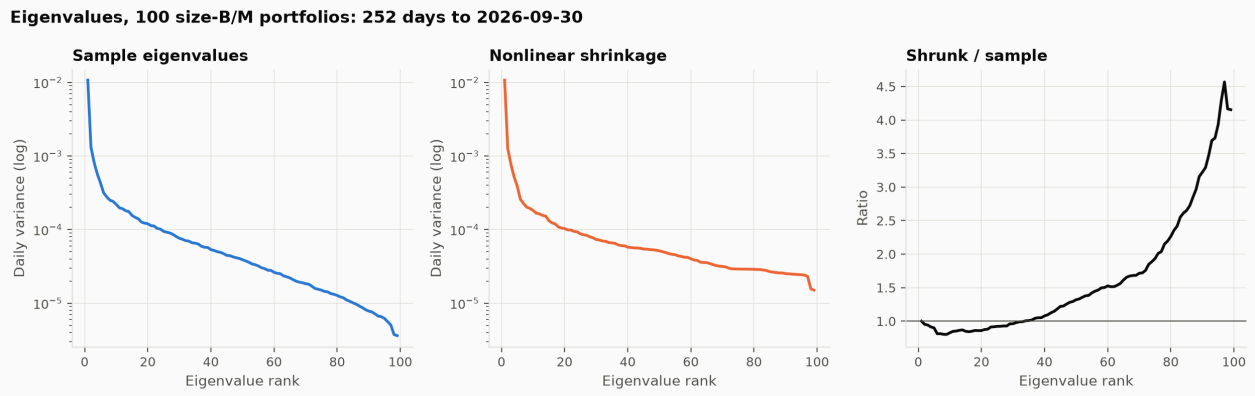


Figure 4: Sample and nonlinear-shrinkage eigenvalues on the last window of 99 portfolios.

On the last 252-day window ending 2026-09-30, nonlinear shrinkage reduces the condition number from 2,958 to 712.

5 Simulation with a known covariance

Real data cannot reveal which forecast was correct. Returns are therefore also drawn from a multivariate Student-t distribution with 5.0 degrees of freedom and a fixed parametric covariance Σ, for N equal to 10, 30, 49 and 100 and T equal to 126, 252, 504 and 1260, with 200 replications per cell. For an estimate H with GMV weights w_H, the true risk $w_H^\top \Sigma\, w_H$ is compared with that of the oracle weights built from Σ itself, $w_\Sigma^\top \Sigma\, w_\Sigma$.

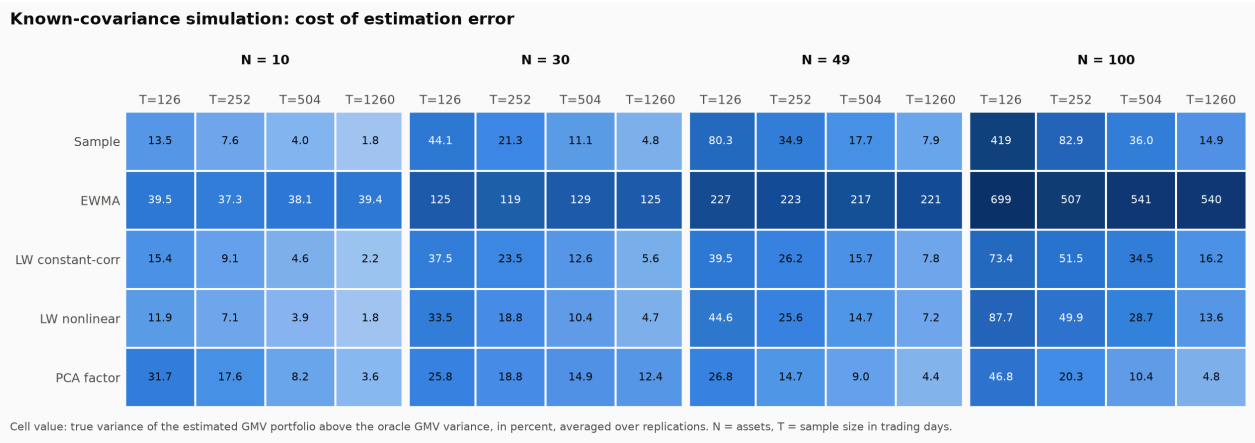


Figure 5: True GMV variance above the oracle in the simulation.

The same values are in `reports/tables/simulation_excess_variance.md`, and their standard errors in `reports/results.json`.

6 Limitations

Portfolio returns are smoother than single-stock returns, so the gaps between estimators understate what a stock-level universe would show; the largest universe, 100 size-B/M portfolios, reaches 0.78 assets per observation. Transaction costs are not charged: turnover is reported beside risk instead. Weights are held fixed within each month so that realized variance isolates forecast quality. The library revised its history when it moved to CRSP's CIZ files in January 2025; every download is hashed, `chr fetch` flags any file whose content changed, and `chr results` rebuilds every number from the revised data.

7 Reproducibility

All code, configuration and tests are in the repository. `uv run chr results` downloads the data, verifies it, runs every stage and regenerates this document and the README from the same `results.json`.

8 References

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